# EC 441: Final Project

Due by December 9, 2024

*Professor David Starobinski*

**Charlie Van Hook & Pree Simphliphan**
cvanhook@bu.edu, prees26@bu.edu

# Part 1: Network Topology and Hub

### Question 1

Based on the tcpdump trace, which hosts receive the ping packet and why?

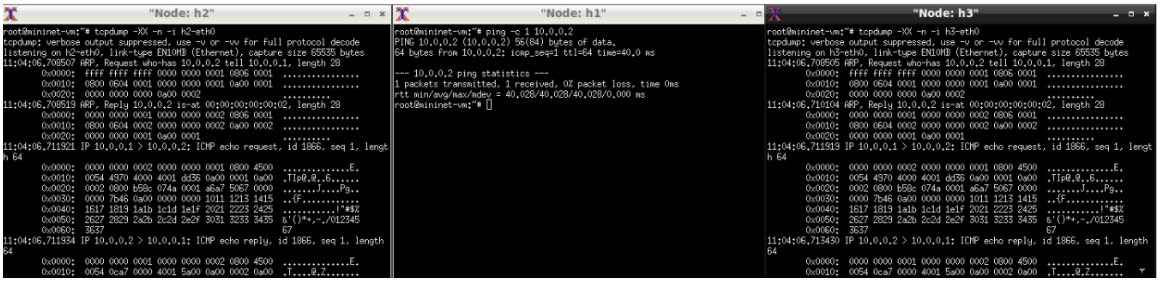


Figure 1: xterm window for each host to view the traffic

<u>Justification:</u> Both hosts (h2 and h3) receive the ping packet because it was broadcast to all hosts connected to the hub.

### Question 2

Consider the tcpdump trace of h3, you should see six packets. Explain what each of these packets is.

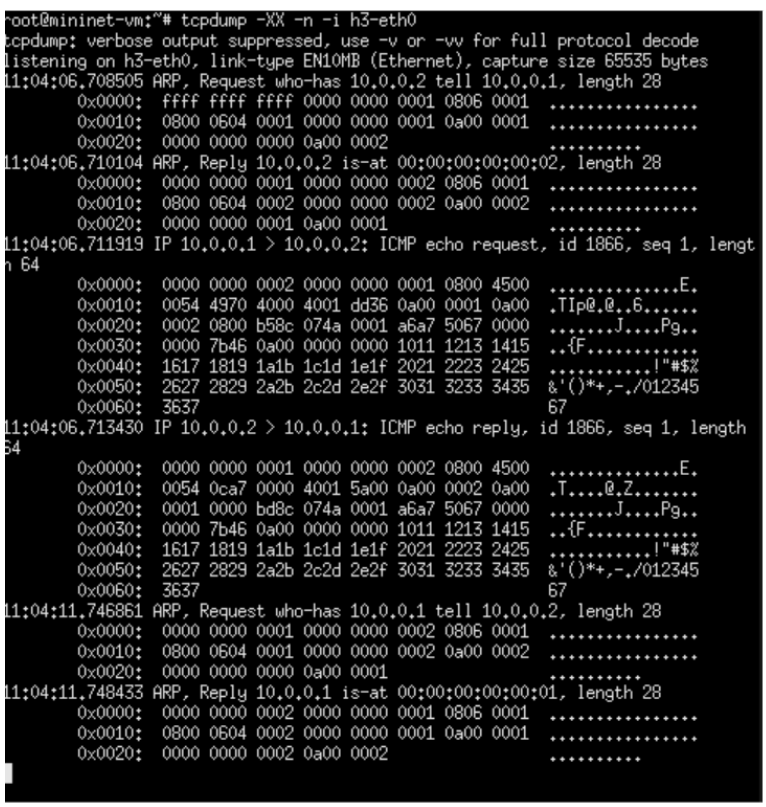


Figure 2: tcpdump trace for h3

Justification: There are six packets because there are three sections, as following; the first section includes the MAC request for IP 10.0.0.2 and its reply; the second section contains the actual ping request and reply packets; the last section includes the MAC request for IP 10.0.0.1 and its reply.

Each section has two packets for the request and reply, totaling six packets.

**Question 3**

From h1 xterm, run ping –c 1 10.0.0.5. Which host will receive the ARP request? Is the request being replied? Will the ping packet be sent and why?

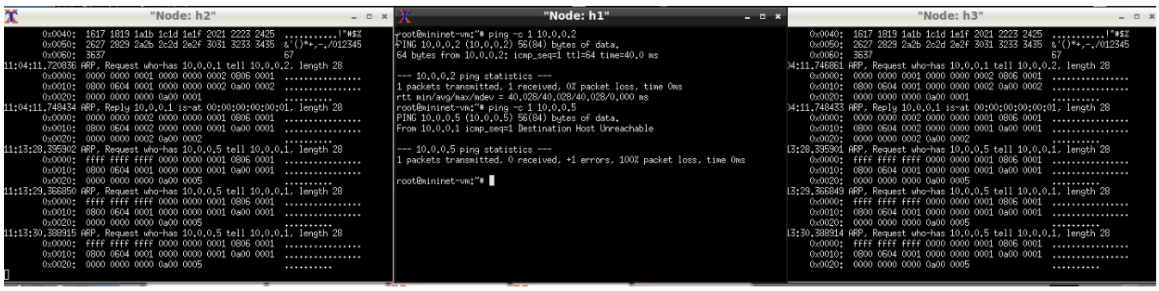


Figure 3: From h1 xterm, run ping –c 1 10.0.0.5.

Justification: Both devices (h2 and h3) receive the ARP request, but there is no reply because none of them are assigned IP 10.0.0.5. Without a reply, the ping packet is not sent since the destination is unknown.

**Question 4**

What is the TCP bandwidth between h1 and h2?

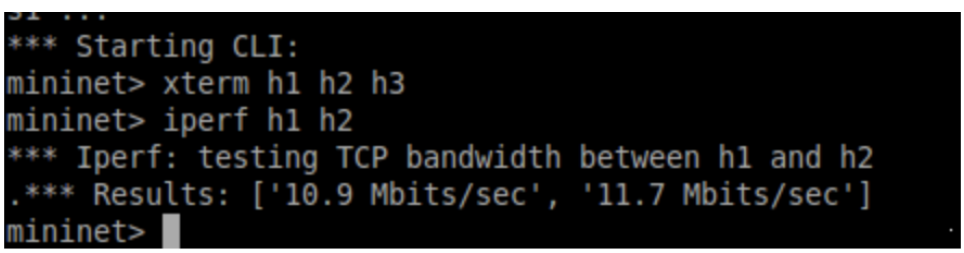


Figure 4: iperf h1 h2 in the Mininet console to test bandwidth between both hosts

Justification: The TCP bandwidth between h1 and h2 is 10.9 Mbits/sec in one direction and 11.7 Mbits/sec in the other direction.

# Part 2: Learning Switch

### Question 5

Based on the tcpdump traces, which hosts receive the ping packet from h1 to h2?

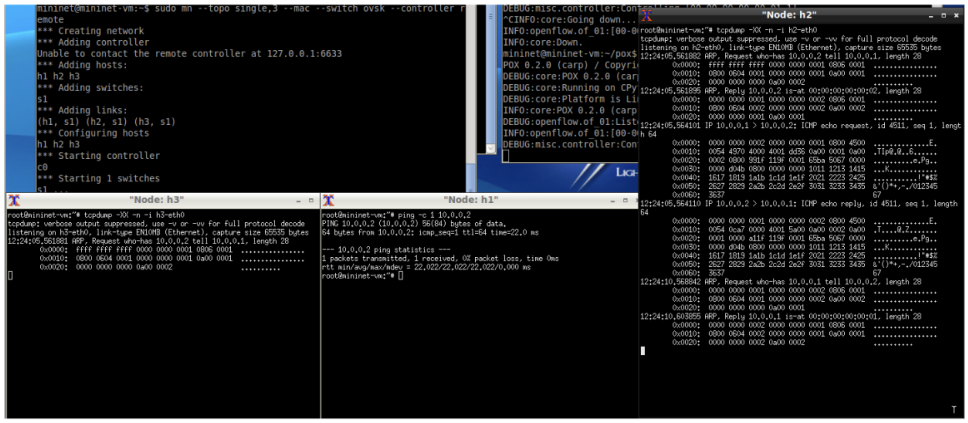


Figure 5: tcpdump trace for h1, h2 and h3

Justification: Only h2 receives the ping packet because the switch has learned the MAC address and port mapping for h1 and h2, allowing it to send the packet directly without broadcasting.

### Question 6

Look at the tcpdump traces of h2 and h3. Why does h2 receive six packets (as in the hub case), while h3 receives only one packet? Make sure to carefully justify your answer

Justification: h2 receives six packets due to the initial broadcast for the MAC address resolution, the actual ping packets, and their respective replies.
The switch already knows the MAC-port mapping for h1 and h2, so it doesn't have to broadcast the reply from h2, therefore only the initial "who is" ARP broadcast will be received by h3.

### Question 7

Question 7: What is the TCP bandwidth between h1 and h2?

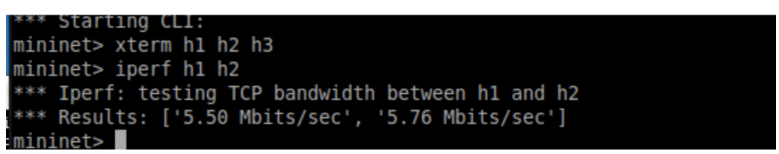


Figure 6: iperf h1 h2 in the Mininet console to test bandwidth between both hosts

Justification: The TCP bandwidth between h1 and h2 is 5.50 Mbits/sec in one direction and 5.76 Mbits/sec in the other direction. The bandwidth is much faster indicating a more efficient data flow with reduced congestion.